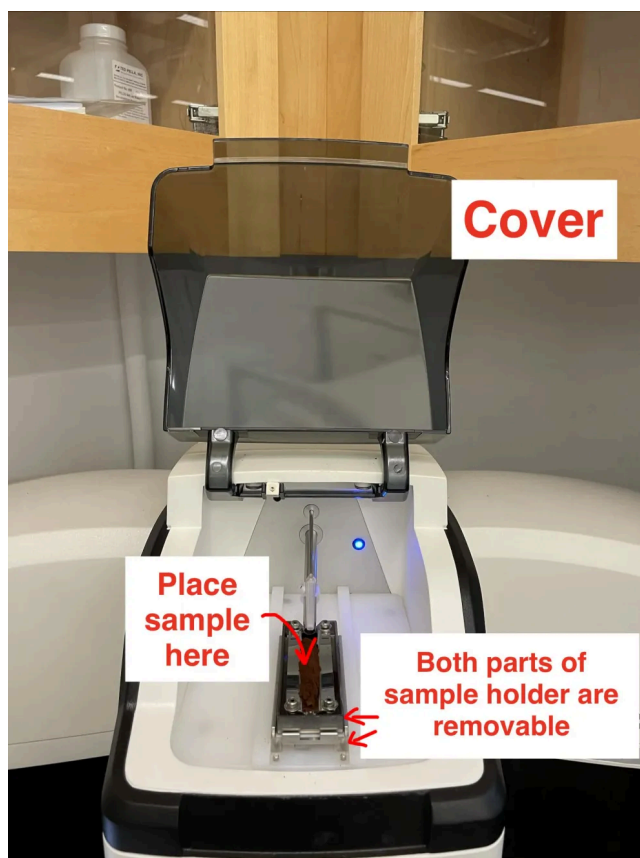


3.000 - Particle Size Analyzer Station

In this rotation, you will grind beans at different grind sizes and perform particle size analysis. *You will need to save the particle size distributions of each of the grind sizes your group tested.*

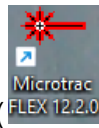
Instructions:

1. Take out your grinders. Each team member should grind their beans at different grind settings spaced apart 4 clicks. For example, a group of three might set grinders so one is 8 clicks, one 12, and one 16.
2. Grind 3 grams of beans each.
3. Open the hood of the particle size analyzer. Add a spoonful of coffee grounds to the sample tray (see example below). Spread evenly, and don't allow the sample beyond the two gray lines.
 - a. If the sample area needs cleaning, you can use a damp paper towel or the powder brush. The sample holders are removable. Ask a TA if you can't figure out how to pull them out.

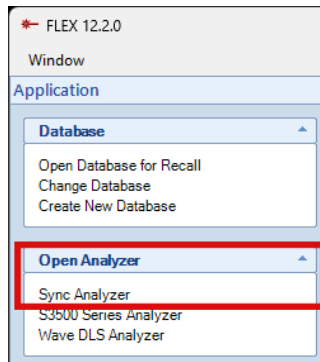


Spoon for scooping sample and brush for cleaning sample tray

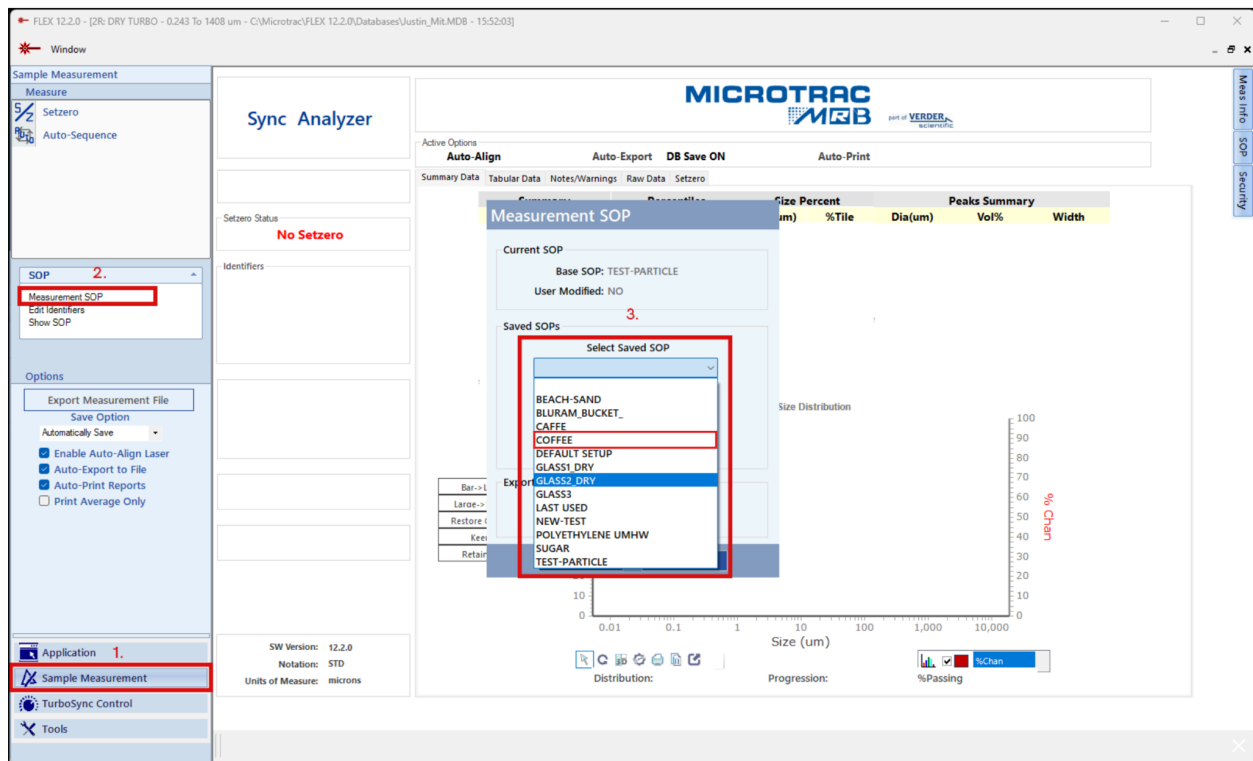
4. Close the gray plastic cover.
5. Log onto the computer with your kerberos.



6. Open Microtrac FLEX (FLEX 12.2.0).
7. From the Open Analyzer menu click Sync Analyzer



8. Click Sample Measurement.
9. Click Measurement SOP, and choose COFFEE.
 - a. The SOP tells the machine how to interpret your particle. For example, the coffee SOP tells the machine that our particles will be irregularly shaped and absorbing (as opposed to transparent or reflective).



10. Click Auto-Sequence ( Auto-Sequence)

11. Name your sample and click OK.

Identifiers Data Entry

Title:

Notes:

Sample IDs:

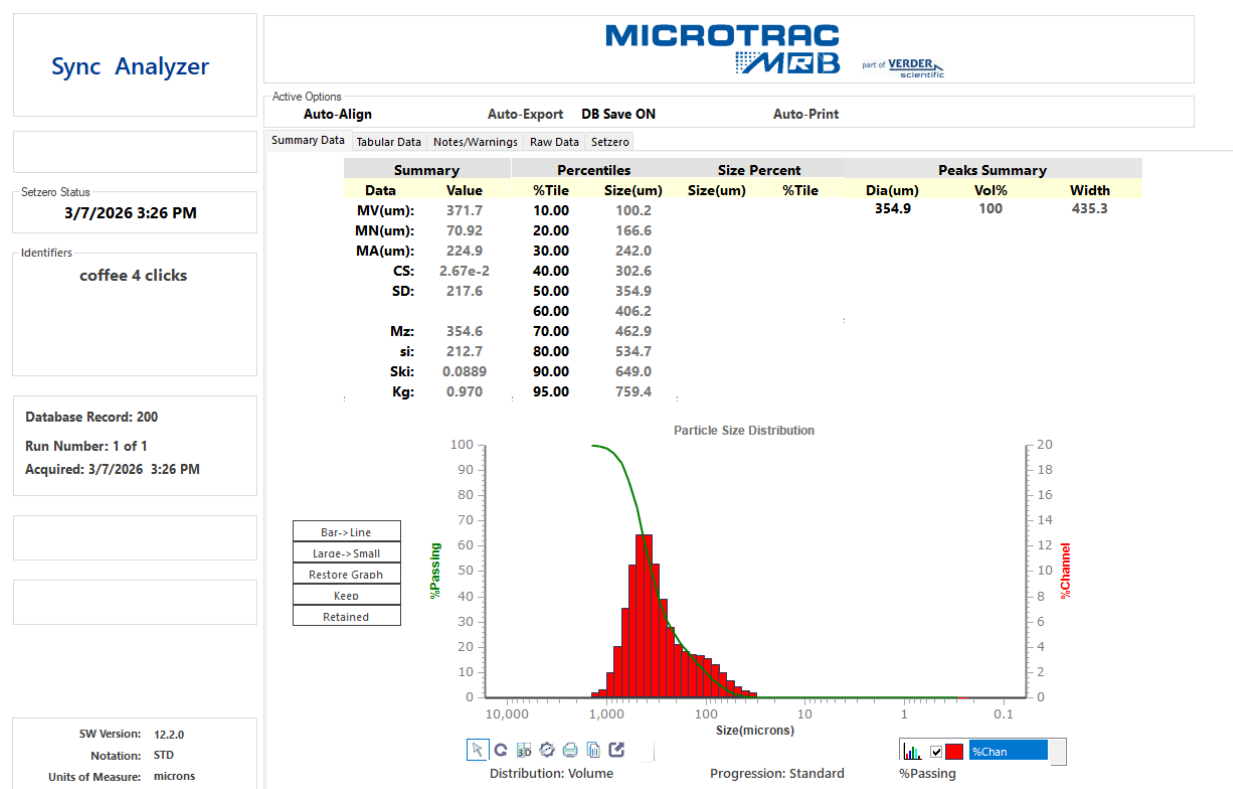
1:

2:

OK

12. Wait for your sample to finish (about a minute).

13. On the auto pop-up, save your results as a pdf or take a screenshot of the results. You will turn in this image on Canvas.



14. Clean out the sample tray with a brush or damp paper towel.

15. Repeat steps 3-12 for each grind size.

- How does the size distribution shift? Is it what you expected?
- How broad is the particle size distribution?
- Do you see any secondary peaks for fines?